

1. JDK and IDE download

Setup and Installation of Java Development Tools

We will go over the installation and setup of various software needed to work with Java in this chapter. The Java Development Kit (JDK) and the Java Virtual Machine (JVM) are required in order to compile and run a Java program.

Software Requirements for Installation

1. Open JDK:

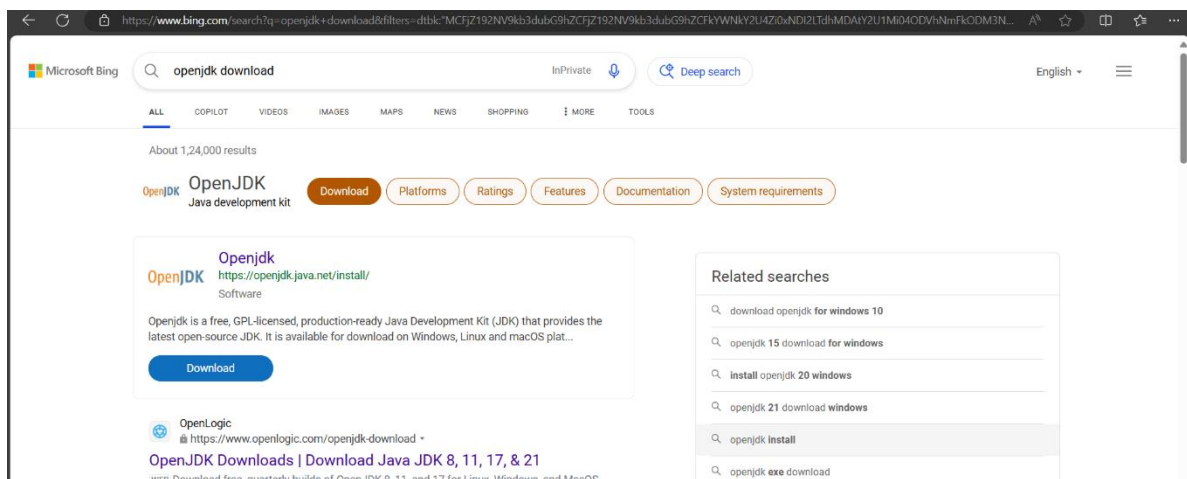
- There are many versions of JDK available on the market. Oracle provides the official JDK, which is licensed and mostly used by big companies because they offer support and charge for it.
- As individuals, we don't typically need this support, so we can use the open-source version of JDK, known as Open JDK.

2. Amazon Corretto:

- This JDK is provided by Amazon.
- There is no significant difference between these JDKs, and multiple vendors provide them, so you can choose any one.

We will install the JDK from the Open JDK website. Currently, JDK 22 is available as the latest version. Java updates its versions every six months with new features. It also has versions designed for long-term support (LTS) such as Java 8, 11, and 17.

We will use Java 20.



https://jdk.java.net/archive/			
Mac/AArch64	64-bit	tar.gz (sha256)	184M
Mac/x64	64-bit	tar.gz (sha256)	186M
Linux/AArch64	64-bit	tar.gz (sha256)	187M
Linux/x64	64-bit	tar.gz (sha256)	189M
		Source	Tags are jdk-20.0.2+9, jdk-20.0.2-ga
20.0.1 (build 20.0.1+9)			
Windows	64-bit	zip (sha256)	188M
Mac/AArch64	64-bit	tar.gz (sha256)	184M
Mac/x64	64-bit	tar.gz (sha256)	186M
Linux/AArch64	64-bit	tar.gz (sha256)	187M
Linux/x64	64-bit	tar.gz (sha256)	189M
		Source	Tags are jdk-20.0.1+9, jdk-20.0.1-ga

Setup for Installation

1. **Choose the JDK file based on your system OS (Linux, Windows, macOS) and download it.**

Apart from the JDK software, we also need Integrated Development Environments (IDEs). Just as we use MS Word or Notepad for writing documents, we need an editor for compiling, running, or debugging programs. An IDE provides an integrated environment with all necessary tools in one place.

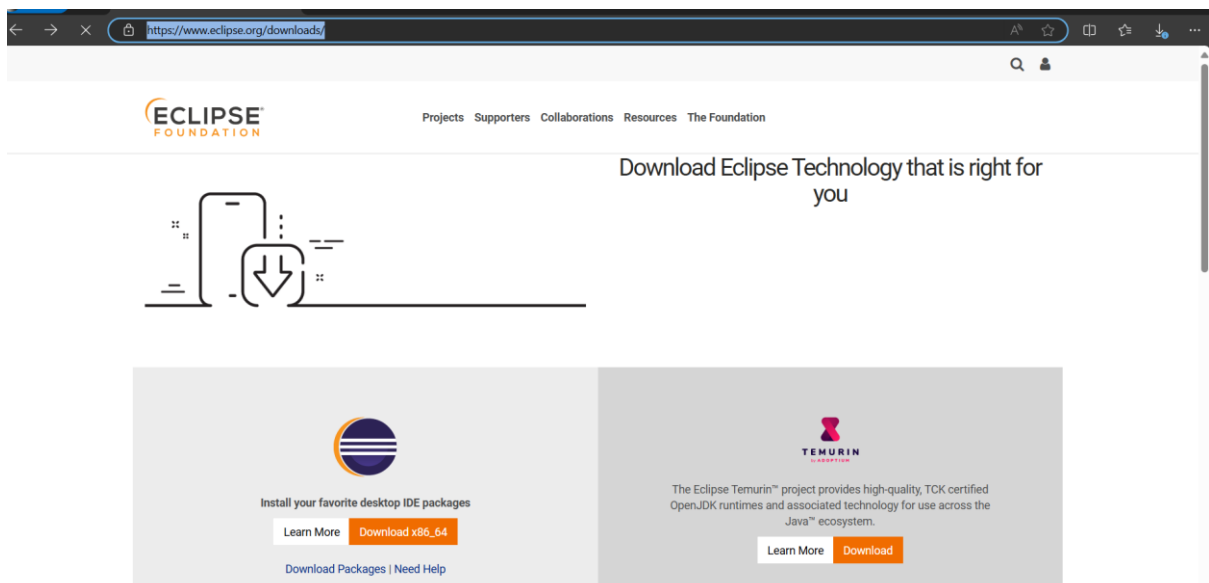
2. **Popular IDEs:**

- **Eclipse:** A popular open-source IDE.
- **Visual Studio Code (VS Code):** Another widely used editor.
- **IntelliJ IDEA:** Known for its powerful features and tools provided.

Steps to Download and Install IDEs

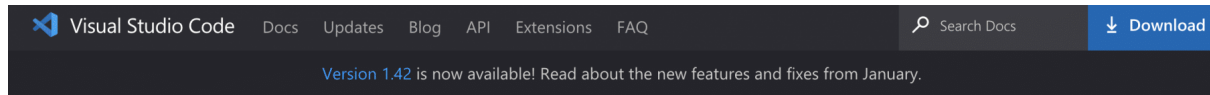
Eclipse:

1. Go to the official [Eclipse website](https://www.eclipse.org/downloads/).
2. Find various versions of Eclipse and choose the Java EE version (Enterprise Edition) to avoid needing to download it later for advanced projects.
3. Download it based on your OS.



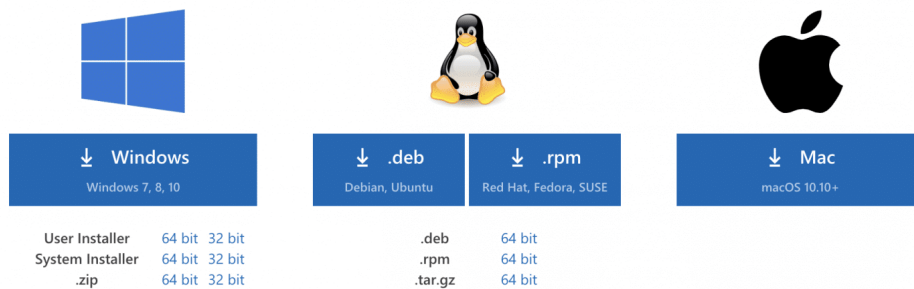
VS Code:

1. Search for "download VS Code" and click on the download link suitable for your OS.
2. You can download Visual Studio code from the URL ["https://code.visualstudio.com/download"](https://code.visualstudio.com/download) by selecting the right platform:
3. The download will start automatically.



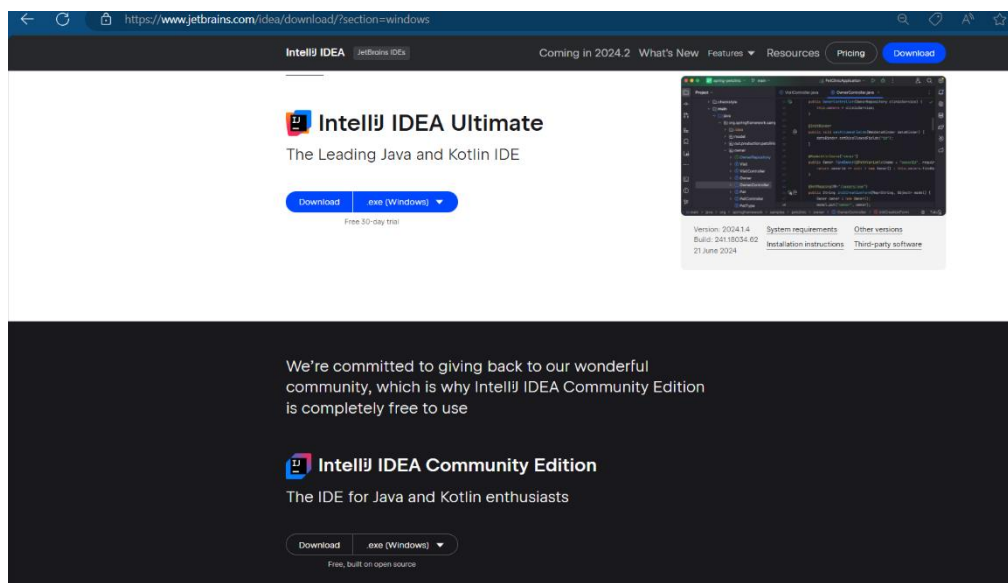
Download Visual Studio Code

Free and built on open source. Integrated Git, debugging and extensions.



IntelliJ IDEA:

1. Navigate to [this link](https://www.jetbrains.com/idea/download/#section=windows) to get the IntelliJ software.
2. Click on the *Download* option.
3. On the next screen, select the platform as Windows and select the edition that you want to download.



4. Download the Community version based on your OS.
5. Save the downloaded file in the desired location.